

Adaptive Multimodal Human–Robot Communication Using a Multi-Agent AI Framework with Safety-Aware Decision Making

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Abstract—Human–Robot Interaction (HRI) is moving toward collaborative environments in which robots are expected to communicate with people in a natural and flexible manner. In practical settings, human instructions are rarely limited to a single communication channel. Speech, gestures, visual cues, and environmental information may occur simultaneously and may provide either complementary or conflicting evidence about the intended task. This paper proposes an adaptive multimodal HRI framework based on a multi-agent artificial intelligence architecture with an explicit safety-verification stage.

The framework divides the interaction process among specialised agents. A Voice Agent interprets spoken instructions, a Gesture Agent analyses hand movements, and a Vision Agent extracts information about objects and the surrounding environment. A Coordinator Agent combines these outputs to estimate the user’s intended action and handle inconsistencies between modalities. A Memory Agent supplies relevant information from previous interactions, while a Task Planner converts the interpreted intent into an ordered set of robot operations. Before execution, a Safety Agent examines the proposed action against environmental and operational constraints.

The resulting architecture is intended to reduce ambiguity in human commands, improve contextual interaction, and prevent unsafe actions from reaching the robot controller. The proposed system can be assessed using intent recognition accuracy, precision, recall, F1-score, task completion rate, response latency, and safety violation rate. Public multimodal datasets together with a purpose-built dataset can be used to evaluate the framework under different interaction conditions.

Index Terms—Human–Robot Interaction, multimodal communication, multi-agent artificial intelligence, large language models, contextual reasoning, robot safety, intent recognition, collaborative robotics.

I. INTRODUCTION

Robots are increasingly being deployed in environments where interaction with humans is part of the task rather than an exceptional event. Examples include healthcare, rehabilitation, manufacturing, logistics, education, domestic assistance, hospitality, and public-service applications. In such settings, successful operation depends not only on the physical capabilities of a robot but also on its ability to interpret human instructions and react appropriately to changing circumstances.

Human communication is inherently multimodal. A person may speak a command while pointing toward an object, using

a particular hand gesture, or interacting with an environment that provides additional contextual information. Consequently, a system that considers only speech or only visual information can lose important evidence required for correct interpretation. Speech may be affected by background noise or ambiguous wording, while gestures can become difficult to recognise because of occlusion, viewpoint, or lighting conditions. These limitations motivate the development of HRI systems that can combine multiple sources of information rather than treating each communication channel independently.

Recent developments in artificial intelligence have created opportunities for building such systems. Speech recognition models can convert spoken instructions into machine-readable representations, computer-vision models can interpret objects and human actions, and large language models can provide higher-level reasoning and task decomposition. Multi-agent architectures provide another useful design principle by assigning specialised functions to individual computational agents instead of forcing one component to perform the complete interaction pipeline.

The framework proposed in this work follows this principle. A Voice Agent handles spoken input, while a Gesture Agent and a Vision Agent provide additional perceptual evidence. Their outputs are forwarded to a Coordinator Agent that performs multimodal reasoning. Instead of making the final decision solely from the current input, a Memory Agent supplies relevant interaction history and user-specific context. The resulting intent is transformed into an executable task by a Task Planner.

Safety is treated as an independent stage rather than being assumed to be implicitly guaranteed by intent recognition. A recognised instruction may still be unsafe because the environment has changed, an obstacle may be present, or the requested manipulation may violate an operational constraint. The Safety Agent therefore evaluates the proposed action before it reaches the robot controller.

The central objective of the proposed framework is to establish an interaction pipeline in which perception, contextual reasoning, task planning, and safety verification are coordinated before physical execution. This design is intended

to improve robustness in situations where human instructions are incomplete, ambiguous, or influenced by environmental changes.

II. LITERATURE REVIEW

Human–Robot Interaction (HRI) has been studied from several perspectives, including multimodal perception, language-based robot control, task planning, adaptive interaction, trust, and safety. In many cases, however, these aspects have been explored separately rather than as parts of one complete interaction system. More recently, researchers have started using large language models (LLMs) and multi-agent architectures to make robots better at reasoning and collaboration. Slezak et al. presented a multi-agent architecture that combines an LLM with retrieval-augmented generation, robotic simulation, and control components. Their results showed that distributing reasoning between different agents can be useful, while physics-based simulation can provide an additional way to validate decisions. At the same time, the study pointed out practical difficulties such as hallucinations, unclear communication between agents, execution delays, and limited experimental conditions [1]. Ok et al. followed a similar direction with a voice-to-action HRI system that connects speech recognition, language-model reasoning, and predefined robot skills. Their system also included safety checks before execution and showed encouraging results in simulation. However, its dependence on a particular robot platform and a fixed set of skills limits how easily it can be applied to other systems, and its real-world evaluation was still limited [2]. Trust has also been considered an important part of human–robot collaboration. Kalluri developed an agent-based simulation framework for studying trust in human–robot teams and reported relationships between robot reliability, communication, transparency, collaboration, and overall team performance [3]. This suggests that effective HRI is not only about understanding what a person says, but also about making the robot’s behaviour clear and predictable.

Other studies have concentrated more closely on individual communication and perception capabilities. Kalusev et al. developed a multi-agent object-detection system using YOLO-based perception together with language-based reporting, communication, and robot-control components on a Raspberry Pi. The system was able to recognise objects locally and provide natural-language reports, although limited computing resources affected the frame rate, processor utilisation, and notification delay [4]. Gesture recognition has traditionally relied on processes such as detection, segmentation, tracking, feature extraction, and classification. These techniques provide a useful basis for understanding human movements, but they generally do not address the larger problem of combining gestures with language-model reasoning and other communication modalities [5]. Voice-controlled robots have also developed considerably, with modern systems linking speech recognition to task planning, robot skills, navigation, and visual information. However, relying mainly on speech can become problematic when a user’s instruction is incomplete or unclear, since useful information may instead be expressed through

gestures or visible features of the environment [14]. Another line of work has focused on risk-aware planning. TAMPURA, for example, incorporates uncertainty and risk into task and motion planning using probabilistic representations and belief-state reasoning. Such approaches are useful when the robot has to make decisions in uncertain environments, but they do not by themselves cover the wider communication process involving speech, gestures, contextual memory, and multimodal interpretation of human intent [15].

Safety has also been explored through multimodal and multi-agent approaches. Multi-agent vision-language systems have been used for identifying potential hazards, showing that different reasoning components can work together for safety-related visual analysis. However, small or partially hidden hazards can still be difficult to detect, and running several model evaluations can increase the computational load [16]. Multimodal datasets are equally important because they provide the data needed to study human activities and evaluate HRI systems. ActionNet, for instance, combines several sensing modalities for human activity understanding and robotic skill learning. Its multimodal structure makes it useful for activity-recognition research, although its focus on kitchen activities means that it does not directly address a complete LLM-based, multi-agent framework for multimodal communication and safety reasoning [6]. Studies on multi-agent collaboration have also reported better coordination and, in some settings, less need for direct human intervention. However, these experiments have often involved relatively small groups of participants or controlled laboratory environments [7]. Adaptive communication has been examined as well, with findings showing that changes in the way a system communicates can influence collaboration quality and how users perceive an intelligent system [8]. Learning-based approaches have extended these capabilities further, including multimodal reinforcement learning systems that use speech, pointing, and haptic or ostensive signals to adapt interaction policies. Even so, these approaches are commonly evaluated within specific tasks or controlled experimental settings [15].

Recent multimodal multi-agent frameworks have begun bringing more of these capabilities together by combining vision-language reasoning, speech, gestures, gaze, locomotion, personality-related information, and long-term memory. These systems show that agent-based architectures can support more personalised forms of interaction, but they also reveal several practical issues, particularly latency, dependence on specific robotic platforms, small-scale evaluations, and the lack of extensive long-term testing in real-world environments [?], [10]. Overall, previous work has established strong foundations for individual HRI capabilities such as speech interaction, gesture recognition, visual perception, language-model reasoning, contextual memory, multi-agent coordination, adaptive communication, and safety-aware planning. The main issue is that these capabilities are still often spread across separate systems or tested only within limited task settings. This leaves an opportunity for a more integrated HRI pipeline in which speech, gestures, and visual information are considered

together before a task is planned, previous interaction context is retained when it is relevant, and the planned action is checked for safety immediately before the robot carries it out.

III. PROBLEM STATEMENT

Current HRI systems frequently specialise in one communication modality or are designed for a narrow task. Such designs become difficult to generalise when humans communicate using several channels simultaneously. A spoken instruction may be incomplete, a gesture may have multiple interpretations, and visual information may contradict the command. Environmental conditions can further increase this uncertainty.

Another issue is the separation between command interpretation and robot safety. Correctly identifying what a user wants does not automatically mean that the requested action should be executed. For example, an otherwise valid movement may become unsafe because an obstacle enters the robot workspace or because the object involved in the task is not suitable for manipulation.

The problem addressed in this research is therefore the design of an adaptive HRI architecture capable of:

- 1) receiving speech, gesture, and visual information simultaneously;
- 2) combining heterogeneous observations into a coherent estimate of human intent;
- 3) incorporating relevant interaction history and user context;
- 4) converting the inferred intent into an executable robot task;
- 5) checking environmental and operational safety constraints before execution; and
- 6) preventing unsafe plans from reaching the robot controller.

The proposed framework will be evaluated using measures related to intent recognition, task completion, latency, safety, and interaction quality. Comparisons with conventional single-modal or non-adaptive HRI pipelines can be used to determine the effect of multimodal fusion and safety-aware decision making.

IV. RESEARCH GAP

The reviewed literature reveals that substantial progress has been achieved in individual areas of HRI, but these capabilities are often implemented as separate modules or evaluated independently.

First, voice-based systems can provide effective spoken control but may lose information available through gestures and visual context. Second, gesture recognition systems can identify human movements but may not provide the higher-level reasoning required to connect a gesture with a language command and an environmental state. Third, multi-agent architectures improve functional decomposition and coordination but do not necessarily provide personalised multimodal communication.

Safety research introduces another distinction. Many safety-aware methods concentrate on navigation, motion planning,

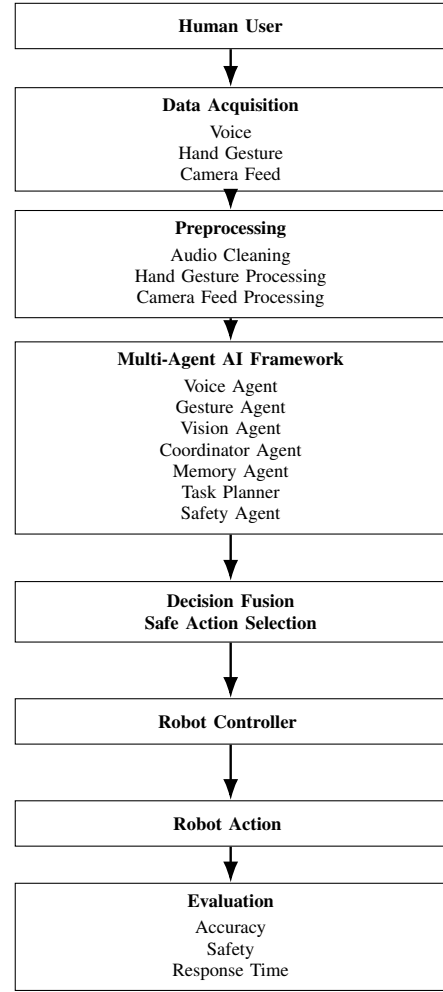


Fig. 1. Proposed pipeline for multimodal multi-agent human-robot communication.

or obstacle avoidance. These are important aspects of robot safety, but they do not necessarily verify whether the interpreted human intention itself is appropriate before execution.

There is consequently a research opportunity for an architecture that treats multimodal perception, contextual memory, intent fusion, task planning, and safety verification as connected stages of one HRI pipeline. Such a framework could also provide a basis for evaluating how different modalities contribute to interaction accuracy under uncertain or changing conditions.

A further gap concerns evaluation outside controlled environments. Several existing approaches have been demonstrated in simulation or restricted laboratory settings. A practical HRI framework requires evaluation across different users, environmental conditions, communication styles, and task types. The proposed research therefore considers both publicly available data and a custom multimodal dataset as possible evaluation resources.

The Voice Agent receives and processes spoken commands. Its output represents the linguistic interpretation of what the user has said. The Gesture Agent examines hand movements

and produces a representation of relevant gestures. The Vision Agent analyses the scene to identify objects, spatial information, and environmental conditions.

The three perception agents provide complementary evidence to the Coordinator Agent. The Coordinator compares and combines these observations before producing an estimate of the user's intended command. This stage is particularly important when one modality is insufficient or when two modalities provide apparently inconsistent information.

V. PROPOSED SYSTEM

The proposed architecture consists of multiple specialised AI agents connected through a central coordination process. The purpose of the architecture is not simply to run several models in parallel, but to assign clear responsibilities to different components of the interaction pipeline.

The Memory Agent maintains information that may remain relevant beyond the current interaction. This can include previous tasks, interaction history, and user-specific preferences. Context from this component is supplied to the reasoning process so that the robot does not have to treat every interaction as completely independent.

Once the intent has been established, the Task Planner generates an ordered set of operations that can be executed by the robot. The plan represents the transition from high-level human intent to low-level robot actions.

Before execution, the proposed plan is examined by the Safety Agent. The agent considers the available environmental information, potential obstacles, object-handling requirements,

TABLE I
MAJOR TECHNOLOGIES USED IN THE FRAMEWORK

Component	Technology
Speech Processing	Whisper / configured speech backend
Gesture Processing	MediaPipe / gesture model
Object Detection	YOLO-based vision pipeline
Agent Orchestration	LangGraph
Programming	Python
Robot Integration	ROS2 interface

and other defined safety constraints. A plan that does not satisfy the required conditions is blocked or returned for reconsideration. A plan that passes the verification stage is forwarded to the Robot Controller.

The conceptual architecture separates perception, reasoning, planning, and safety into distinct components while allowing them to operate together as one integrated system.

VI. SYSTEM METHODOLOGY

The proposed framework follows a staged decision pipeline in which information from different human communication channels is first converted into structured observations and then progressively refined into a robot action. Instead of allowing an individual modality to directly control the robot, the system separates perception, multimodal interpretation, contextual reasoning, planning, safety assessment, and execution. The workflow is implemented using specialized agents coordinated through a state-based processing pipeline. This organization allows uncertainty or conflicts detected at an earlier stage to be considered before an action reaches the robot controller.

A. Multimodal Input Processing

The system accepts three primary sources of interaction information: spoken commands, hand gestures, and camera-based environmental observations. These inputs are handled separately at the initial stage because each modality provides different information about the user's intention and the surrounding environment.

Speech is processed through the Voice Agent, which converts an audio command into a textual representation and extracts information relevant to the requested task. Depending on the configured input mode, the speech component can work with a speech-recognition backend or a text input used for software-level testing. The resulting information is represented in a structured form so that later stages can use the detected action, target, spatial information, and associated confidence.

Hand gestures are processed by the Gesture Agent. The agent identifies the gesture and records relevant directional information when a pointing gesture is present. Directional gestures are particularly useful when a spoken instruction does not uniquely identify the intended object. Rather than directly interpreting a gesture as a robot command, the system retains the gesture as an observation for subsequent multimodal reasoning.

The Vision Agent processes camera observations to obtain information about objects and their positions within the scene.

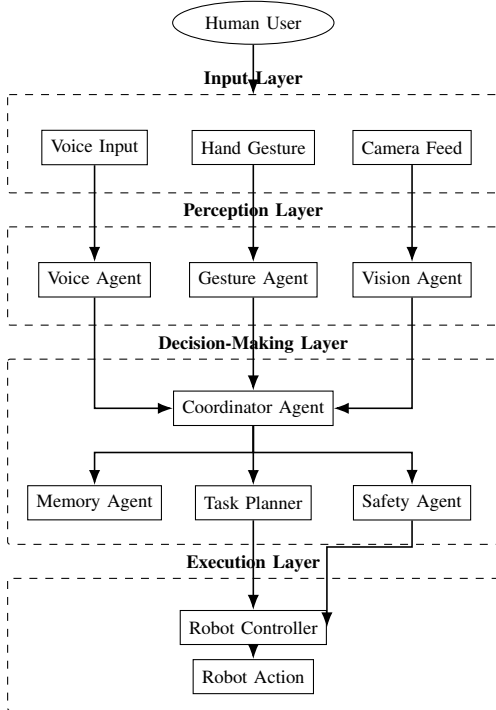


Fig. 2. Proposed multimodal multi-agent architecture for adaptive human-robot communication and safety-aware decision making.

Detected objects are represented using their class labels, confidence values, bounding boxes, and spatial sectors. This representation provides environmental information that can be associated with the linguistic and gestural observations generated by the other agents.

Consequently, the input stage converts heterogeneous sensor information into structured observations that can be passed to the perception and decision-making stages.

B. Coordinator Agent

The Coordinator Agent forms the central decision stage of the proposed framework. It is implemented using LangGraph as a state-based workflow for coordinating multimodal observations and task interpretation. The Coordinator receives structured outputs from the Voice, Gesture, and Vision Agents and processes them through a sequence of validation, multimodal fusion, target grounding, conflict analysis, and task interpretation steps.

The Coordinator does not directly execute robot actions. Instead, it produces a structured representation of the interpreted task that can be passed to the Memory Agent and Task Planner. This separation allows uncertainty in perception or multimodal interpretation to be represented before the task reaches the planning and safety stages.

The conceptual Coordinator workflow can be represented as
Input Validation → Multimodal Fusion → Target Grounding
→ Conflict Analysis → Task Interpretation.

LangGraph serves as the orchestration mechanism for this workflow, while the Coordinator Agent represents the application-level multimodal reasoning component. Thus, LangGraph is not treated as an additional HRI agent.

C. Intent Fusion

The Coordinator Agent combines the structured outputs produced by the Voice, Gesture, and Vision Agents to determine whether the available observations support a consistent interpretation of the user’s request. The fusion process is implemented as a state-based workflow using LangGraph, allowing intermediate decisions and interaction states to be passed between successive processing stages.

The fusion process considers speech, gesture, and visual information jointly rather than treating each modality as an independent command source. Speech generally provides the semantic action and, when explicitly stated, the target object. Gesture information can provide spatial or directional evidence, while vision supplies information about objects and their positions in the current scene.

A distinction is made between explicit and indirect target references. When the linguistic input explicitly specifies an object, the identified object is retained as the primary target and gesture or visual information is used for verification and spatial consistency checking. A conflicting gesture therefore does not automatically replace an explicitly specified target with another visual candidate.

For indirect references such as “it”, “this”, or “that object”, the Coordinator performs multimodal target grounding. Candidate objects are evaluated using available visual confidence, spatial agreement with the gesture, gesture confidence, and relevant contextual information. Object-detection confidence is therefore treated as one component of target grounding rather than as the sole target-selection criterion.

The Coordinator can represent the resulting grounding condition using states such as high-confidence grounding, low-confidence grounding, ambiguity, target-not-found, and spatial conflict. These states allow uncertain interpretations to remain explicit instead of being treated as unconditionally executable commands.

For the implemented fusion configuration, modality confidence values can be represented using

$$C_f = w_v C_v + w_{vis} C_{vis} + w_g C_g, \quad (1)$$

where C_v , C_{vis} , and C_g denote the available voice, vision, and gesture confidence values, respectively, and w_v , w_{vis} , and w_g represent their corresponding weights.

The resulting fused representation is forwarded to the contextual reasoning and task-planning stages only after the available evidence has been evaluated. When the evidence is insufficient or contradictory, the interaction remains in an unresolved state for further handling rather than being directly converted into a robot action.

1) *Confidence-Aware Target Grounding*: Consider an indirect command such as “Pick it up.” If the user points toward the right side of the scene and the Vision Agent detects a potted plant in that region, the gesture provides spatial evidence for associating the pronoun with the detected object. The confidence assigned to the visual detection is nevertheless retained as part of the grounding decision.

For example, a grounding state can be represented as

Gesture	POINT_RIGHT
Gesture confidence	0.92
Candidate object	Potted plant
Spatial agreement	TRUE
Visual confidence	0.49
Grounding state	LOW_CONFIDENCE

In this situation, the potted plant can remain a candidate target because its spatial location agrees with the gesture. However, the relatively low visual confidence prevents the system from treating the grounding evidence as equivalent to an explicitly identified and confidently detected target. The resulting state can therefore be passed to the safety stage for further verification or confirmation.

D. Contextual Reasoning

The Memory Agent provides interaction history that may be relevant to the interpretation of a current instruction. This is particularly useful when a user refers to an object or action indirectly across successive interactions.

Referential grounding can use both current multimodal evidence and relevant interaction history. The Coordinator

prioritizes current perceptual information for the present scene, while the Memory Agent supplies previously established context when the current instruction depends on an earlier interaction.

For example, a user may first identify an object and subsequently issue a command referring to “it” without repeating the object’s name. The Memory Agent can provide the previously established task context to the Coordinator, which can then consider that information together with the current gesture and visual observations.

Context is treated as supporting evidence rather than as an unconditional source of decisions. If the available history does not provide sufficient evidence, the system retains an unresolved state instead of arbitrarily selecting a target. This separation prevents stored context from overriding current scene information when the available evidence is inconsistent.

E. Task Planning

Following successful intent interpretation and contextual resolution, the Task Planner transforms the high-level task into an ordered sequence of robot actions. The planner operates on the structured task representation produced by the preceding stages rather than directly processing raw speech, gesture, or camera data.

The generated plan depends on the type of task and the information available about its target. Supported task categories include movement-related actions, object inspection, pick-and-place operations, and stopping. Target-dependent tasks are checked for the presence of the information required to construct a meaningful action sequence.

The planner decomposes a high-level request into primitive operations that can be understood by the robot controller. Spatial information obtained during perception and fusion can be incorporated into the plan so that directional movement or target alignment can occur before the main operation. Each plan is represented as an ordered collection of actions together with the associated parameters and execution information.

This intermediate planning representation separates task-level reasoning from low-level robot control. As a result, the controller does not have to interpret natural-language instructions directly; it receives a structured plan that has already passed through the interpretation stages.

F. Safety Verification

Safety verification is positioned between task planning and robot execution so that generating a valid task plan does not by itself authorize the robot to perform the corresponding action.

The Safety Agent evaluates the planned action together with environmental information and the confidence associated with the preceding interpretation stages. Deterministic checks include emergency-stop conditions, proximity constraints, sensor validity, insufficient perception confidence, missing targets, and unresolved interaction states.

The safety stage also considers the confidence of referential grounding for target-dependent actions. When a target has been inferred indirectly from a deictic expression and the

TABLE II
SAFETY HANDLING FOR DIFFERENT INTERACTION CONDITIONS

Condition	Safety Handling
High-confidence target with no detected safety violation	Eligible for execution
Low-confidence referential grounding	Further verification or human confirmation
Ambiguous target	Further verification or human confirmation
Target not found	Block execution
Explicit target with spatial conflict	Retain explicit target and require conflict handling
Emergency-stop condition	Block execution
Unsafe proximity	Block execution
Invalid sensor condition	Block execution

available multimodal evidence is insufficient to establish the target reliably, the action is not treated as an unconditionally executable command. Instead, the system can retain the uncertain state for further verification or human intervention.

The Safety Agent also incorporates proximity hysteresis to reduce unstable transitions when environmental measurements fluctuate near a configured safety boundary. Contextual risk reasoning can provide additional information, but it does not replace deterministic safety constraints.

The safety stage therefore acts as a decision gate between planning and execution. A simplified representation is

$$E = \begin{cases} 1, & \text{if the planned action satisfies the required conditions,} \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

Here, $E = 1$ indicates that the action is eligible to proceed toward execution, whereas $E = 0$ indicates that the action must be withheld or redirected for further handling.

The separation between task planning and safety verification ensures that an interpreted intention is not automatically treated as an executable command. This is particularly important for indirect target references, conflicting multimodal observations, and changing environmental conditions.

G. Robot Execution

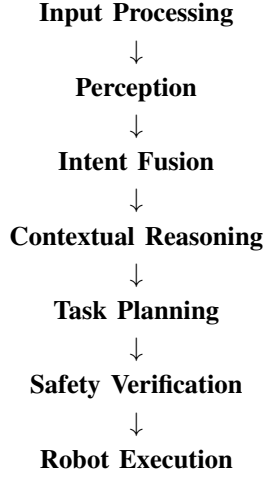
The Robot Controller forms the final stage of the decision pipeline. It receives the action sequence generated by the Task Planner together with the corresponding safety decision and converts the approved operations into robot-level commands.

The controller supports primitive operations required by the framework, including directional movement, stopping, picking, placing, and inspection. The separation between planning and control allows the same high-level task representation to be mapped to the appropriate robot interface without changing the preceding reasoning stages.

The implementation also provides a software-level robot interface for testing the decision pipeline and includes an interface intended for ROS2-based robotic integration. Therefore, the methodology separates the AI decision process from the hardware-specific execution layer. This makes it possible to evaluate the interaction logic independently before connecting the framework to a physical robot platform.

Before execution, the controller verifies that the action has received the required safety approval. If execution encounters an error, the controller can issue a stop operation before reporting the failure to the rest of the system. Execution information can subsequently be retained as part of the interaction history.

The complete processing flow is summarized as follows:



This ordering establishes a controlled path from human communication to robot action, with multimodal interpretation, contextual resolution, planning, and safety assessment occurring before execution. The resulting architecture allows the individual agents to perform specialized functions while the overall state-based workflow coordinates their outputs into a unified interaction process.

VII. EXPERIMENTAL SETUP

The implemented framework was evaluated at the software-integration level using the complete multimodal HRI pipeline. The evaluation focuses on the interaction among the Voice, Gesture, Vision, Coordinator, Memory, Task Planner, Safety, and Robot Controller components rather than evaluating each perception model independently. The experiments were conducted through the integrated HRI pipeline, which coordinates the individual agents through a state-based workflow.

A. Evaluation Environment

The evaluation uses structured inputs representing speech, gesture, visual observations, and environmental safety conditions. This configuration enables controlled reproduction of interaction scenarios while examining the behavior of the complete decision-making pipeline.

The implemented configuration uses confidence thresholds for perception and intent processing, predefined modality weights for multimodal fusion, contextual memory for referential resolution, and configurable safety conditions. The Robot Controller is evaluated through the software pipeline, and the reported processing latency represents the execution time of the integrated software workflow rather than physical robot movement time.

B. Scenario-Based Test Design

A set of 21 hand-crafted Human–Robot Interaction scenarios was used to examine different interaction conditions. The scenarios cover ordinary commands as well as cases involving ambiguity, multimodal information, conflicting observations, unavailable targets, contextual references, and safety-related conditions.

The test set includes voice-only interactions, combinations of voice and visual information, and cases involving voice, vision, and gesture simultaneously. Additional scenarios examine conflicting spatial information, target-not-found conditions, ambiguous references, low-confidence inputs, invalid commands, stop-related behavior, environmental safety conditions, and multi-turn interactions.

The multi-turn scenarios examine whether information established during an earlier interaction can be reused when a subsequent command contains an indirect reference. For example, an object identified during one interaction can be referenced in a later command without explicitly repeating its name.

Each test case contains predefined expected conditions for the relevant task state, target, action, and, where applicable, safety state. The actual output generated by the integrated pipeline is compared with the corresponding expected outcome to determine whether the scenario satisfies the specified test condition.

C. Evaluation Procedure

For each scenario, structured representations of the required input modalities are supplied to the integrated pipeline. When a modality is not required for a particular scenario, the corresponding input is omitted. Visual scenarios provide detected-object information together with spatial information, while gesture scenarios provide the recognized gesture and its associated direction.

Environmental parameters are supplied for scenarios requiring safety evaluation. These parameters can include human distance, obstacle distance, emergency-stop status, and sensor validity. The resulting state is then processed by the integrated HRI pipeline.

The processing time is recorded immediately before and after pipeline execution. The end-to-end latency is calculated as

$$T_{\text{latency}} = (t_{\text{end}} - t_{\text{start}}) \times 1000, \quad (3)$$

where t_{start} and t_{end} represent the beginning and completion times of the pipeline execution, respectively.

The resulting task state is compared with the expected state specified for the corresponding test case. For each scenario, the evaluation records the actual task status, target, action, safety decision, and processing latency.

D. Evaluation Metrics

The evaluation primarily considers system-level decision behavior rather than isolated perception-model benchmarks. The

following measures are used to characterize the implemented system.

Task completion rate: This measure represents the proportion of evaluated scenarios for which the observed system output satisfies the predefined expected outcome and is calculated as

$$R_{\text{task}} = \frac{N_{\text{passed}}}{N_{\text{total}}} \times 100. \quad (4)$$

Intent and fusion performance: The evaluation examines whether the system produces the expected task state for valid commands, ambiguous inputs, target-not-found conditions, and multimodal conflicts.

Context resolution: Multi-turn scenarios are used to examine whether previously stored interaction information can support the interpretation of subsequent referential commands.

Safety decision performance: For scenarios containing an explicitly specified safety condition, the safety state produced by the system is compared with the corresponding expected state.

End-to-end latency: The elapsed processing time from submission of the test state to generation of the final software-level decision is recorded for every scenario. Aggregate latency statistics are obtained from the recorded execution times.

Model-specific measures such as Word Error Rate (WER), mean Average Precision (mAP), and Intersection over Union (IoU) are not used as primary evaluation measures in this experiment because the current scenario-based evaluation does not provide sufficiently annotated, environment-specific ground-truth data for these metrics.

E. Test Categories

The evaluated scenarios cover several types of system behavior:

- 1) *Basic command interpretation* – evaluation of direct task requests.
- 2) *Multimodal grounding* – association of speech information with visual and gestural observations.
- 3) *Modality conflict* – examination of cases in which different input channels provide inconsistent information.
- 4) *Target availability* – evaluation of commands referring to unavailable or undetected objects.
- 5) *Confidence handling* – examination of incomplete or low-confidence observations.
- 6) *Contextual interaction* – evaluation of multi-turn commands and referential expressions.
- 7) *Safety conditions* – examination of stop-related, proximity-related, and sensor-related states.
- 8) *Controller-level decisions* – verification that the final task state is passed correctly toward the robot execution stage.

This categorization allows the evaluation to examine the complete decision pipeline rather than restricting the analysis to an individual agent.

F. Reproducibility

The evaluation is implemented as a repeatable software procedure. The test scenarios are stored in a structured test-case configuration, while the evaluation procedure constructs the required modality inputs, invokes the integrated pipeline, records execution latency, compares the generated state with the predefined expected outcome, and stores the resulting observations.

The evaluation produces individual records for each scenario together with aggregate measurements. The same scenario set can therefore be executed again after modifications to the Coordinator, Memory, Task Planner, Safety Agent, or Robot Controller, allowing changes in system behavior to be examined under a consistent evaluation procedure.

G. Confidence-Aware Grounding Results

The additional grounding scenarios were evaluated to examine how the framework handles indirect target references and conflicting multimodal evidence. The analysis distinguishes high-confidence grounding from low-confidence and ambiguous cases rather than treating all successful target associations as equivalent.

The results are reported separately from the overall scenario pass rate because grounding confidence represents an intermediate decision state rather than a standalone perception accuracy measure. In particular, low-confidence and ambiguous cases are examined to determine whether uncertainty is retained before task execution.

VIII. RESULTS AND DISCUSSION

The implemented framework was evaluated using the 21 scenario-based test cases described in Section VII. The experiments were designed to examine the behavior of the complete multimodal pipeline under different combinations of speech, gesture, visual information, contextual references, and safety-related conditions. The reported results therefore represent system-level behavior of the implemented pipeline and should not be interpreted as independent benchmark scores for the underlying speech, gesture, or object-detection models.

A. Overall Scenario Performance

The evaluation produced a successful outcome for all 21 test scenarios according to the current scenario-level evaluation criterion. Thus, the observed scenario pass rate was

$$R_{\text{scenario}} = \frac{21}{21} \times 100 = 100\%. \quad (5)$$

The result indicates that every test case was completed without being classified as a failed scenario by the evaluation procedure. The tested cases included single-modality commands, multimodal interactions, target resolution, contextual references, conflicting observations, and safety-related inputs.

However, the value should be interpreted specifically as a *scenario pass rate*. It does not represent a general-purpose accuracy measure for speech recognition, gesture recognition, object detection, or real-world robot task execution. The test

TABLE III
END-TO-END SOFTWARE PIPELINE LATENCY

Metric	Latency (ms)
Mean	6.96
Median	4.55
Minimum	2.98
Maximum	48.07
Standard deviation	9.36

set consists of a limited number of manually constructed scenarios and therefore cannot establish generalisation to unseen environments or users.

B. Multimodal Interaction Results

The evaluated scenarios demonstrate that the implemented pipeline can process different combinations of communication modalities within a common decision workflow. Voice-only cases provide the basic command interpretation path, while multimodal cases introduce visual and gestural information for additional target or spatial context.

The multimodal scenarios also examine situations in which information from different sources does not directly agree. In such cases, the Coordinator Agent represents the interaction state explicitly rather than assuming that every input combination corresponds to a valid executable task. This behaviour allows ambiguous or conflicting observations to be identified before task planning and execution.

The contextual interaction scenarios further demonstrate the role of the Memory Agent. In multi-turn cases, information established during an earlier interaction can be used when a subsequent command contains an indirect reference. This provides a mechanism for maintaining continuity between successive interaction steps rather than interpreting every command independently.

C. Execution Latency

End-to-end processing time was measured for each of the 21 scenarios. The observed latency statistics are summarized in Table III.

The mean processing time was approximately 6.96 ms, while the median was 4.55 ms. The difference between these values is influenced by a small number of comparatively longer executions, with the maximum observed latency reaching 48.07 ms. The standard deviation of 9.36 ms indicates that the processing time was not completely uniform across the evaluated scenarios.

These measurements represent the execution time of the implemented software decision pipeline. They do not include physical robot motion, actuator response, communication delay to external hardware, or the time required for a robot to complete a planned physical task. Consequently, the measured latency should not be interpreted as complete human-to-robot response time in a deployed robotic system.

D. Safety-Related Observations

Safety-related scenarios were included to examine the interaction between task interpretation, environmental conditions,

and the Safety Agent. The framework provides explicit safety states that can influence whether a planned task is allowed to proceed toward execution.

The current scenario evaluation successfully processed the safety-related test cases according to the evaluation procedure. However, the safety results require careful interpretation because the present test configuration contains cases in which the expected safety state does not fully correspond to the descriptive condition of the scenario. For this reason, a separate numerical safety-accuracy value is not reported here.

This distinction is important because a scenario-level pass result alone is insufficient to establish that the safety mechanism achieves a particular safety accuracy. A stronger safety evaluation would require a formally defined safety ground truth, independently verified hazardous conditions, and a larger set of positive and negative safety cases.

E. Discussion

The results provide an initial validation of the integrated decision pipeline. The successful completion of the evaluated scenarios indicates that the implemented agents can be connected into a common workflow covering multimodal input interpretation, contextual reasoning, task planning, safety assessment, and controller-level decision making.

The latency measurements also indicate that the software-level processing path can complete the evaluated scenarios within a relatively short computational interval. Nevertheless, these measurements should be considered only as an indication of pipeline processing cost because the experiment does not include physical robot movement or hardware communication delays.

The scenario-based evaluation also reveals the importance of separating system-level validation from individual model benchmarking. Although the current evaluation can determine whether a predefined interaction scenario reaches the expected system state, it does not provide sufficient evidence for claims concerning the general accuracy of the underlying speech, gesture, or object-detection models. Such measurements require independent benchmark datasets with appropriate ground-truth annotations.

Similarly, the current results do not establish performance under large variations in users, accents, lighting conditions, gesture styles, object arrangements, or dynamic physical environments. These factors should therefore be examined in subsequent experiments before making broader claims about real-world deployment.

Overall, the current results support the feasibility of the implemented multimodal agent workflow at the software integration level while also identifying the need for larger-scale and independently validated experiments for perception accuracy, safety performance, and physical robot execution.

IX. LIMITATIONS AND FUTURE WORK

Although the implemented framework demonstrates the feasibility of integrating multimodal perception, agent-based

reasoning, contextual memory, task planning, and safety verification within a common HRI pipeline, the present study has several limitations. These limitations mainly arise from the scale of the current evaluation, the software-level nature of the experiments, and the absence of large-scale ground-truth data for independent perception benchmarking.

A. Limitations

The first limitation concerns the size and nature of the evaluation set. The current experimental study uses 21 manually constructed interaction scenarios. Although these scenarios cover different combinations of voice, gesture, vision, contextual interaction, and safety conditions, they represent a controlled evaluation rather than a statistically large collection of interactions from multiple users and environments. Therefore, the observed scenario pass rate should not be interpreted as a measure of generalisation to unseen users or real-world operating conditions.

A second limitation is the absence of independent benchmark evaluation for the individual perception components. The current experiment evaluates the integrated decision pipeline rather than independently measuring speech recognition, gesture recognition, or object detection performance. In particular, Word Error Rate (WER), mean Average Precision (mAP), and Intersection over Union (IoU) are not reported because the current evaluation does not contain sufficiently annotated, environment-specific ground-truth data for these measurements. Consequently, the present results cannot be used to establish the comparative performance of the individual perception models.

Another limitation is that the reported latency represents software-level pipeline processing. The measurement includes the processing performed by the integrated decision workflow but does not represent the complete physical response time of a deployed robot. Actuator delay, motor response, robot movement, hardware communication, and other physical execution effects are outside the scope of the current latency measurement.

The present study also does not include a large-scale human-subject evaluation. As a result, factors such as user preference, interaction workload, communication naturalness, trust, and usability have not been quantitatively assessed. Such factors are important when evaluating a human-robot communication framework intended for repeated interaction.

Safety evaluation also requires further validation. The current test scenarios provide an initial software-level assessment of safety-related decision handling, but they do not constitute a comprehensive safety certification or physical safety assessment. A larger set of independently defined hazardous and non-hazardous situations is required to quantify false-safe and false-unsafe decisions under controlled conditions.

Finally, the current framework has been evaluated primarily through predefined scenarios. Real operating environments can introduce variations in lighting, background objects, speech conditions, gesture visibility, sensor quality, and unexpected user behaviour. The effect of these variations on multimodal

fusion and downstream planning therefore remains to be established experimentally.

B. Future Work

A further direction is confidence calibration for multimodal target grounding. The current framework represents uncertainty using perception, gesture, spatial, and contextual evidence, but the effectiveness of the selected confidence thresholds requires evaluation using larger annotated interaction datasets. Future experiments can investigate adaptive threshold selection and calibration using diverse users, objects, environments, and interaction styles.

Future work can also examine longer sequences of deictic references and multimodal conflicts. Such experiments would help determine how grounding confidence changes when the scene changes between successive commands or when current perception disagrees with previously established context.

X. CONCLUSION

This work presented an adaptive multimodal Human-Robot Interaction framework that combines speech, hand gestures, visual perception, contextual memory, task planning, and safety verification within a multi-agent architecture. The framework separates perception from decision making and execution, allowing information from different communication channels to be processed and combined before a robot action is considered for execution.

The implemented system was evaluated through 21 scenario-based interaction tests covering single-modality commands, multimodal inputs, contextual references, conflicting observations, target-related conditions, and safety-related situations. The current evaluation recorded 21 successful scenarios according to the predefined scenario-level evaluation criteria. The software pipeline showed a mean processing latency of approximately 6.96 ms, with a median of 4.55 ms and a maximum of 48.07 ms across the evaluated scenarios. These measurements demonstrate the feasibility of the implemented decision pipeline at the software-integration level.

The results should be interpreted within the scope of the current evaluation. The experiments do not establish general perception accuracy, physical robot response time, or comprehensive real-world safety performance. In particular, the current study does not include large-scale user evaluation, independent WER or mAP benchmarking, or extensive physical-robot trials. These limitations provide clear directions for further validation.

Future work will therefore focus on larger and more diverse multimodal datasets, independent evaluation of the perception components, physical robot integration, controlled ablation studies, extended safety testing, and user-based HRI evaluation. Such experiments will provide a stronger basis for assessing the robustness and generalisation of the proposed framework in practical human-robot interaction environments.

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